

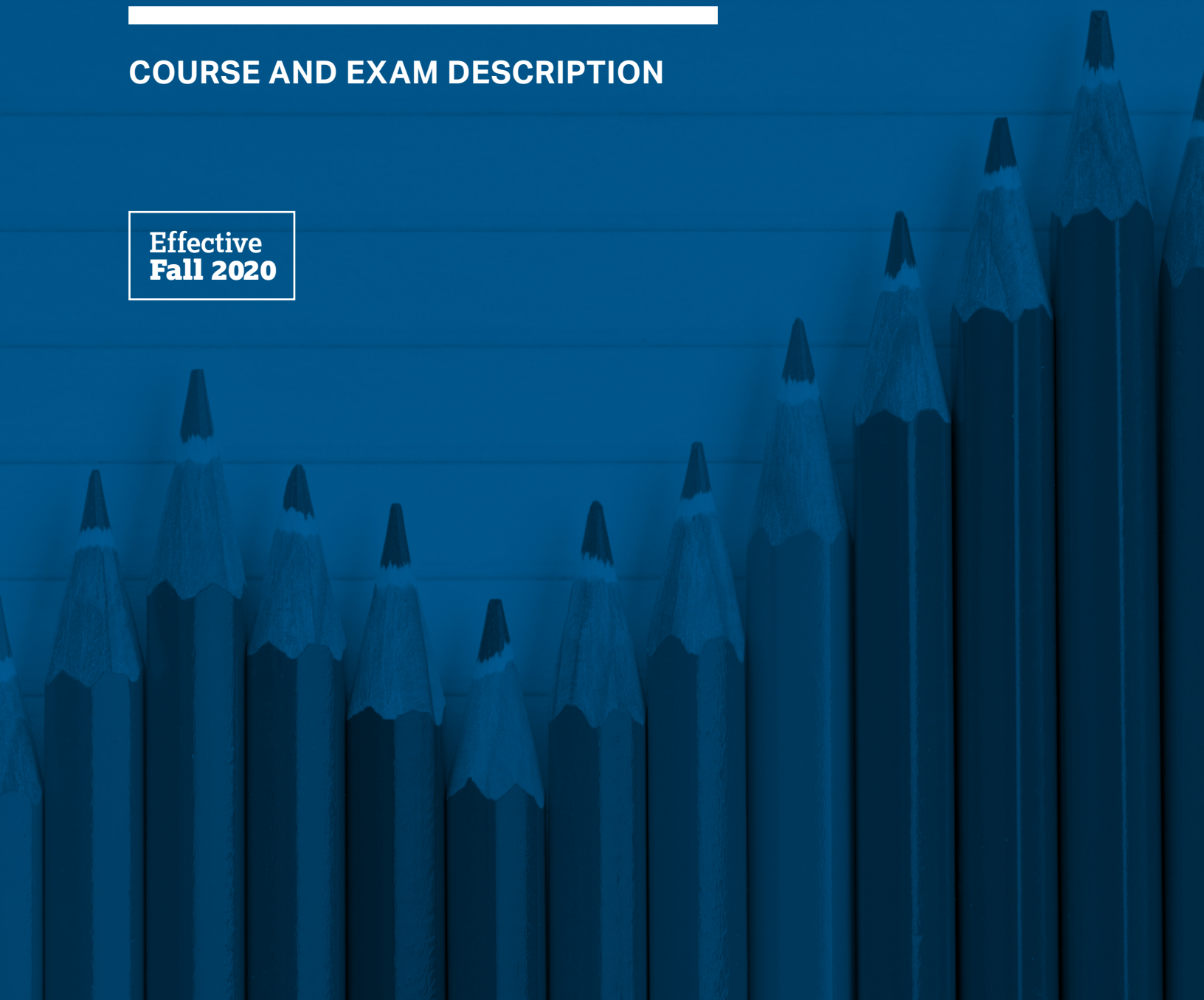
INCLUDES

- ✓ Course framework
- ✓ Instructional section
- ✓ Sample exam questions
- ✓ Classroom poster

AP[®] Statistics

COURSE AND EXAM DESCRIPTION

Effective
Fall 2020



Formulas for AP Statistics

I. Descriptive Statistics

$$\begin{aligned}\bar{x} &= \frac{1}{n} \sum x_i = \frac{\sum x_i}{n} & s_x &= \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \\ \hat{y} &= a + bx & \bar{y} &= a + b\bar{x} \\ r &= \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) & b &= r \frac{s_y}{s_x}\end{aligned}$$

II. Probability and Distributions

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability Distribution	Mean	Standard Deviation
Discrete random variable, X	$\mu_X = E(X) = \sum x_i \cdot P(x_i)$	$\sigma_X = \sqrt{\sum (x_i - \mu_X)^2 \cdot P(x_i)}$
If X has a binomial distribution with parameters n and p , then: $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$ where $x = 0, 1, 2, 3, \dots, n$	$\mu_X = np$	$\sigma_X = \sqrt{np(1-p)}$
If X has a geometric distribution with parameter p , then: $P(X = x) = (1-p)^{x-1} p$ where $x = 1, 2, 3, \dots$	$\mu_X = \frac{1}{p}$	$\sigma_X = \frac{\sqrt{1-p}}{p}$

III. Sampling Distributions and Inferential Statistics

Standardized test statistic: $\frac{\text{statistic} - \text{parameter}}{\text{standard error of the statistic}}$
Confidence interval: $\text{statistic} \pm (\text{critical value})(\text{standard error of statistic})$
Chi-square statistic: $\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$

III. Sampling Distributions and Inferential Statistics (*continued*)

Sampling distributions for proportions:

Random Variable	Parameters of Sampling Distribution	Standard Error* of Sample Statistic
For one population: $\hat{p}$	$\mu_{\hat{p}} = p$ $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$	$s_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
For two populations: $\hat{p}_1 - \hat{p}_2$	$\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$ $\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$	$s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ When $p_1 = p_2$ is assumed: $s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\hat{p}_c(1-\hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$ where $\hat{p}_c = \frac{X_1 + X_2}{n_1 + n_2}$

Sampling distributions for means:

Random Variable	Parameters of Sampling Distribution	Standard Error* of Sample Statistic
For one population: $\bar{X}$	$\mu_{\bar{X}} = \mu$ $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$	$s_{\bar{X}} = \frac{s}{\sqrt{n}}$
For two populations: $\bar{X}_1 - \bar{X}_2$	$\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2$ $\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	$s_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

Sampling distributions for simple linear regression:

Random Variable	Parameters of Sampling Distribution	Standard Error* of Sample Statistic
For slope: b	$\mu_b = \beta$ $\sigma_b = \frac{\sigma}{\sigma_x \sqrt{n}}$ where $\sigma_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$	$s_b = \frac{s}{s_x \sqrt{n-1}}$ where $s = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n-2}}$ and $s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$

*Standard deviation is a measurement of variability from the theoretical population. Standard error is the estimate of the standard deviation. If the standard deviation of the statistic is assumed to be known, then the standard deviation should be used instead of the standard error.

Tables for AP Statistics

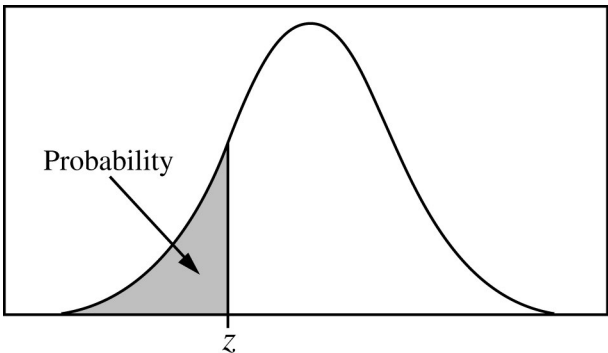


Table entry for z is the probability lying below z .

Table A Standard Normal Probabilities

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559

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Table A Standard Normal Probabilities (continued)

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

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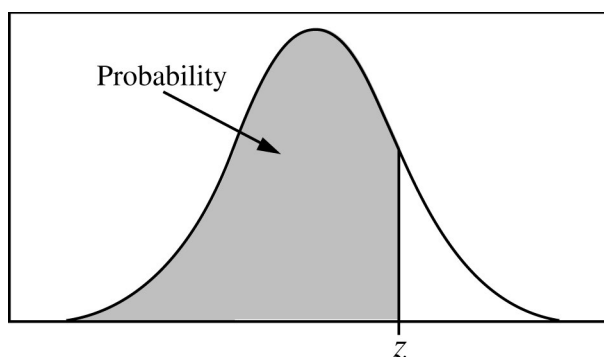


Table entry for z is the probability lying below z .

Table A Standard Normal Probabilities (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936

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Table A **Standard Normal Probabilities (continued)**

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

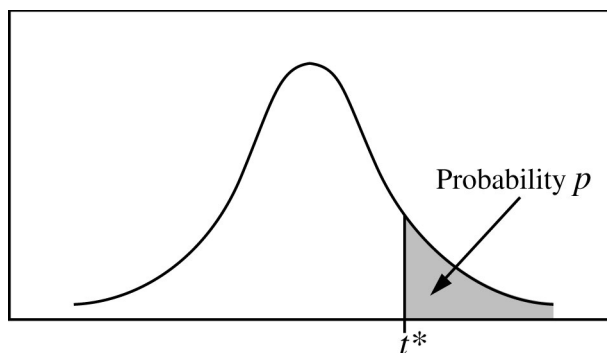


Table entry for p and C is the point t^* with probability p lying above it and probability C lying between $-t^*$ and t^* .

Table B t Distribution Critical Values

df	Tail Probability p											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850

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Table B *t* Distribution Critical Values (continued)

<i>df</i>	Tail Probability <i>p</i>											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
21	.686	.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	.684	.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	.681	.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	.679	.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	.679	.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	.678	.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	.677	.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	.675	.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	.674	.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291
	50%	60%	70%	80%	90%	95%	96%	98%	99%	99.5%	99.8%	99.9%
Confidence level C												

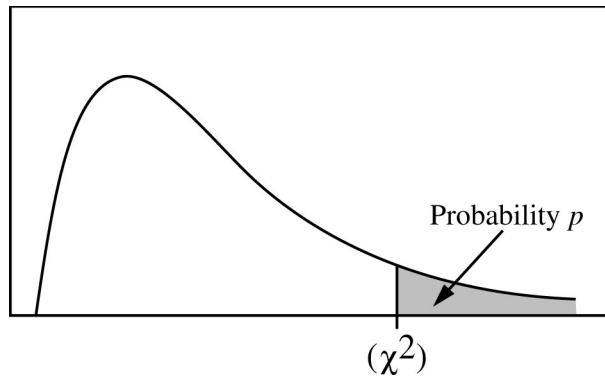


Table entry for p is the point (χ^2) with probability p lying above it.

Table C χ^2 Critical Values

df	Tail probability p											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.32	1.64	2.07	2.71	3.84	5.02	5.41	6.63	7.88	9.14	10.83	12.12
2	2.77	3.22	3.79	4.61	5.99	7.38	7.82	9.21	10.60	11.98	13.82	15.20
3	4.11	4.64	5.32	6.25	7.81	9.35	9.84	11.34	12.84	14.32	16.27	17.73
4	5.39	5.99	6.74	7.78	9.49	11.14	11.67	13.28	14.86	16.42	18.47	20.00
5	6.63	7.29	8.12	9.24	11.07	12.83	13.39	15.09	16.75	18.39	20.51	22.11
6	7.84	8.56	9.45	10.64	12.59	14.45	15.03	16.81	18.55	20.25	22.46	24.10
7	9.04	9.80	10.75	12.02	14.07	16.01	16.62	18.48	20.28	22.04	24.32	26.02
8	10.22	11.03	12.03	13.36	15.51	17.53	18.17	20.09	21.95	23.77	26.12	27.87
9	11.39	12.24	13.29	14.68	16.92	19.02	19.68	21.67	23.59	25.46	27.88	29.67
10	12.55	13.44	14.53	15.99	18.31	20.48	21.16	23.21	25.19	27.11	29.59	31.42
11	13.70	14.63	15.77	17.28	19.68	21.92	22.62	24.72	26.76	28.73	31.26	33.14
12	14.85	15.81	16.99	18.55	21.03	23.34	24.05	26.22	28.30	30.32	32.91	34.82
13	15.98	16.98	18.20	19.81	22.36	24.74	25.47	27.69	29.82	31.88	34.53	36.48
14	17.12	18.15	19.41	21.06	23.68	26.12	26.87	29.14	31.32	33.43	36.12	38.11
15	18.25	19.31	20.60	22.31	25.00	27.49	28.26	30.58	32.80	34.95	37.70	39.72
16	19.37	20.47	21.79	23.54	26.30	28.85	29.63	32.00	34.27	36.46	39.25	41.31
17	20.49	21.61	22.98	24.77	27.59	30.19	31.00	33.41	35.72	37.95	40.79	42.88
18	21.60	22.76	24.16	25.99	28.87	31.53	32.35	34.81	37.16	39.42	42.31	44.43
19	22.72	23.90	25.33	27.20	30.14	32.85	33.69	36.19	38.58	40.88	43.82	45.97
20	23.83	25.04	26.50	28.41	31.41	34.17	35.02	37.57	40.00	42.34	45.31	47.50

continued on next page

Table C χ^2 Critical Values (continued)

<i>df</i>	Tail probability <i>p</i>											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
21	24.93	26.17	27.66	29.62	32.67	35.48	36.34	38.93	41.40	43.78	46.80	49.01
22	26.04	27.30	28.82	30.81	33.92	36.78	37.66	40.29	42.80	45.20	48.27	50.51
23	27.14	28.43	29.98	32.01	35.17	38.08	38.97	41.64	44.18	46.62	49.73	52.00
24	28.24	29.55	31.13	33.20	36.42	39.36	40.27	42.98	45.56	48.03	51.18	53.48
25	29.34	30.68	32.28	34.38	37.65	40.65	41.57	44.31	46.93	49.44	52.62	54.95
26	30.43	31.79	33.43	35.56	38.89	41.92	42.86	45.64	48.29	50.83	54.05	56.41
27	31.53	32.91	34.57	36.74	40.11	43.19	44.14	46.96	49.64	52.22	55.48	57.86
28	32.62	34.03	35.71	37.92	41.34	44.46	45.42	48.28	50.99	53.59	56.89	59.30
29	33.71	35.14	36.85	39.09	42.56	45.72	46.69	49.59	52.34	54.97	58.30	60.73
30	34.80	36.25	37.99	40.26	43.77	46.98	47.96	50.89	53.67	56.33	59.70	62.16
40	45.62	47.27	49.24	51.81	55.76	59.34	60.44	63.69	66.77	69.70	73.40	76.09
50	56.33	58.16	60.35	63.17	67.50	71.42	72.61	76.15	79.49	82.66	86.66	89.56
60	66.98	68.97	71.34	74.40	79.08	83.30	84.58	88.38	91.95	95.34	99.61	102.7
80	88.13	90.41	93.11	96.58	101.9	106.6	108.1	112.3	116.3	120.1	124.8	128.3
100	109.1	111.7	114.7	118.5	124.3	129.6	131.1	135.8	140.2	144.3	149.4	153.2

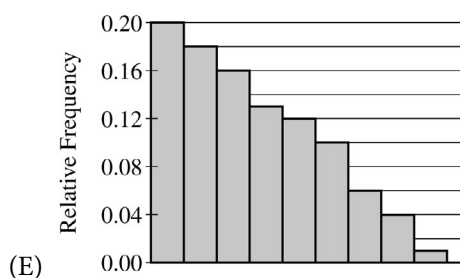
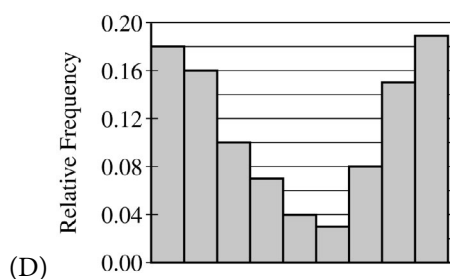
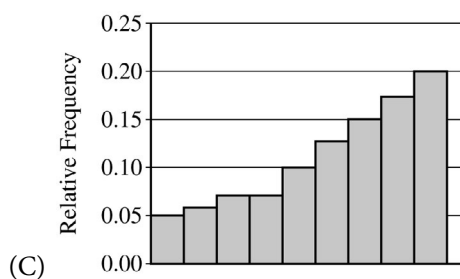
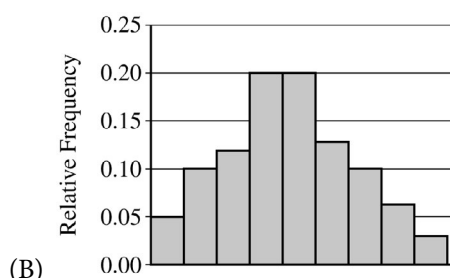
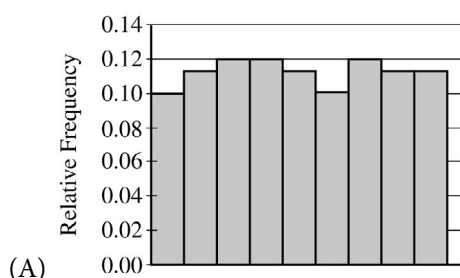
Sample Exam Questions

The sample exam questions that follow illustrate the relationship between the course framework and AP Statistics Exam and serve as examples of the types of questions that appear on the exam. After the sample questions is a table that shows which skill, learning objective(s), and unit each question relates to. The table also provides the answers to the multiple-choice questions.

Section I: Multiple-Choice

The following are examples of the kinds of multiple-choice questions found on the exam.

1. Which of the following histograms has a shape that is approximately uniform?



2. A recent study reported that 45 percent of adults in the United States now get all their news online. A random sample of 8 adults in the United States will be selected. What is the probability that 6 of the selected adults get all their news online?

(A) $\binom{6}{2}(0.45)^8(0.55)^2$

(B) $\binom{6}{2}(0.45)^6(0.55)^2$

(C) $\binom{8}{6}(0.45)^2(0.55)^8$

(D) $\binom{8}{6}(0.45)^6(0.55)^2$

(E) $\binom{8}{6}(0.45)^8(0.55)^6$

3. In 2011, 17 percent of a random sample of 200 adults in the United States indicated that they consumed at least 3 pounds of bacon that year. In 2016, 25 percent of a random sample of 600 adults in the United States indicated that they consumed at least 3 pounds of bacon that year.

Assuming all conditions for inference are met, which of the following is the most appropriate test statistic to use to investigate whether the proportion of all adults in the United States who consume at least 3 pounds of bacon in 2016 is different from that of 2011?

(A) $z = \frac{0.17 - 0.25}{\sqrt{\frac{(0.17)(0.83)}{200} + \frac{(0.25)(0.75)}{600}}}$

(B) $z = \frac{0.17 - 0.25}{\sqrt{\frac{(0.17)(0.83)}{600} + \frac{(0.25)(0.75)}{200}}}$

(C) $z = \frac{0.17 - 0.25}{\sqrt{(0.21)(0.79)\left(\frac{1}{200} + \frac{1}{600}\right)}}$

(D) $z = \frac{0.17 - 0.25}{\sqrt{(0.23)(0.77)\left(\frac{1}{200} + \frac{1}{600}\right)}}$

(E) $z = \frac{0.17 - 0.25}{\sqrt{(0.50)(0.50)\left(\frac{1}{200} + \frac{1}{600}\right)}}$

4. In a recent year, the distribution of age for senators in the United States Senate was unimodal and roughly symmetric with mean 65 years and standard deviation 10.6 years. Consider a simulation with 200 trials in which, for each trial, a random sample of 5 senators' ages is selected and the mean age calculated.

Which of the following best describes the distribution of the 200 sample mean ages?

- (A) Approximately normal with mean 65 years and standard deviation 10.6 years
- (B) Approximately normal with mean 65 years and standard deviation $\frac{10.6}{\sqrt{5}}$ years
- (C) Approximately normal with mean 65 years and standard deviation $\frac{10.6}{\sqrt{200}}$ years
- (D) Approximately uniform with mean 65 years and standard deviation $\frac{10.6}{\sqrt{5}}$ years
- (E) Approximately uniform with mean 65 years and standard deviation $\frac{10.6}{\sqrt{200}}$ years

5. An athletic director believes that more than 50 percent of the students in a certain school district exercise at least 3.5 hours per week. To investigate the belief, the director selected a random sample of 40 students in the school district and found that 60 percent of the students in the sample exercise at least 3.5 hours per week. Let p represent the proportion of all students in the school district who exercise at least 3.5 hours per week.

Which of the following is the most appropriate alternative hypothesis to test the director's claim?

- (A) $H_a : p = 0.5$
- (B) $H_a : p > 0.5$
- (C) $H_a : p < 0.5$
- (D) $H_a : p = 0.6$
- (E) $H_a : p > 0.6$

6. For a certain population of penguins, the distribution of weight is approximately normal with mean 15.1 kilograms (kg) and standard deviation 2.2 kg. Approximately what percent of the penguins from the population have a weight between 13.0 kg and 16.5 kg?

- (A) 17%
- (B) 34%
- (C) 57%
- (D) 68%
- (E) 74%

7. A researcher collected data on the latitude, in degrees north of the equator, and the average low temperature, in degrees Fahrenheit, for a random sample of cities in Europe. The data were used to create the following equation for the least-squares regression line.

$$\text{predicted average low temperature} = 65.5 - 0.70(\text{latitude})$$

Which of the following is the best interpretation of the slope of the line?

- (A) For each one degree north of the equator increase, the predicted average low temperature increases on average by 0.70 degree Fahrenheit.
 - (B) For each one degree north of the equator increase, the predicted average low temperature decreases on average by 0.70 degree Fahrenheit.
 - (C) For each 0.7 degree north of the equator increase, the predicted average low temperature decreases on average by 1 degree Fahrenheit.
 - (D) For each one degree Fahrenheit increase in average low temperature, the predicted latitude increases on average by 0.70 degree north of the equator.
 - (E) For each one degree Fahrenheit increase in average low temperature, the predicted latitude decreases on average by 0.70 degree north of the equator.
8. An independent polling agency was hired to track the preferences of registered voters in a district for an upcoming election. The polling agency divided the district into twenty regions and believes that the regions are similar to one another in their composition. The agency then randomly selected two of the regions and surveyed all registered voters in both regions.

Which of the following best describes the sampling method used by the polling agency?

- (A) Convenience sampling
 - (B) Simple random sampling
 - (C) Stratified random sampling
 - (D) Systematic sampling
 - (E) Cluster sampling
9. A sports physician conducted a study to investigate whether there is an association between running experience and the occurrence of a certain sport injury for marathon runners while training for a marathon. Data were collected on a random sample of 51 marathon runners. Each runner from the sample was categorized by running experience (low, medium, high) and whether or not the runner experienced the sport injury while training for a marathon. The conditions for inference were met, and a χ^2 test statistic of approximately 8.12 was calculated.

Which of the following describes the p -value of the test?

- (A) $p\text{-value} > 0.25$
- (B) $0.10 < p\text{-value} < 0.25$
- (C) $0.05 < p\text{-value} < 0.10$
- (D) $0.01 < p\text{-value} < 0.05$
- (E) $p\text{-value} < 0.01$

10. An agricultural engineer selected a random sample of 30 farms in the United States to construct a 95 percent confidence interval for the mean size, in acres, of farms in the United States. The resulting interval was (367, 558).

Which of the following is an appropriate interpretation of the 95 percent confidence level?

- (A) Approximately 95% of the farm sizes in the sample are between 367 acres and 558 acres.
 - (B) Approximately 95% of all farm sizes in the United States are between 367 acres and 558 acres.
 - (C) Approximately 95% of all random samples of size 30 from the population will have a mean farm size between 367 acres and 558 acres.
 - (D) Approximately 95% of all random samples of size 30 from the population will produce intervals that contain the mean size of farms in the United States.
 - (E) Approximately 95% of all random samples of size 30 from the population will produce intervals that contain the sample mean.
11. An experiment was designed to investigate the relationship between the dosage of a certain migraine medication and the amount of time until relief from the headache was experienced. Four dosages in milligrams (mg)—2.5, 5.0, 7.5, and 10—of the medication were used. The 20 participants in the experiment were known to experience migraines. Each participant was randomly assigned one of the four dosages. When the participants experienced a migraine, they took the assigned medication and recorded the number of minutes it took to experience relief from the headache. The mean number of minutes it took each group to experience relief was compared.
- What were the experimental units in the experiment?
- (A) The number of minutes to experience relief
 - (B) The mean number of minutes it took each group to experience relief
 - (C) The four dosages
 - (D) The 20 participants
 - (E) The relationship between the dosage and the amount of time
12. The recommended dietary allowance of iron for women under the age of 51 is 18 milligrams (mg) per day. A medical researcher studying women living in a certain region suspected that the women were getting less than the daily allowance of iron, on average. The researcher took a random sample of women under the age of 51 from the region and measured their daily iron intakes. The following hypotheses were tested at the significance level of $\alpha = 0.05$ for the population mean μ of the daily iron intake for women in the region.

$$H_0 : \mu = 18$$

$$H_a : \mu < 18$$

All conditions for inference were met, and the resulting p -value was 0.031.

Which of the following is an appropriate conclusion?

- (A) The p -value is less than α , and the null hypothesis should be rejected. There is convincing statistical evidence that the mean daily intake of iron for women in the region is less than the recommended 18 mg.
- (B) The p -value is less than α , and the null hypothesis should be rejected. There is not convincing statistical evidence that the mean daily intake of iron for women in the region is less than the recommended 18 mg.
- (C) The p -value is less than α , and the null hypothesis should not be rejected. There is not convincing statistical evidence that the mean daily intake of iron for women in the region is less than the recommended 18 mg.
- (D) The p -value is greater than α , and the null hypothesis should be rejected. There is convincing statistical evidence that the mean daily intake of iron for women in the region is less than the recommended 18 mg.
- (E) The p -value is greater than α , and the null hypothesis should not be rejected. There is not convincing statistical evidence that the mean daily intake of iron for women in the region is less than the recommended 18 mg.

13. At a small company, 40 percent of the employees are classified as junior level, and 32 percent of the employees work in department A. Of those classified as junior level, 55 percent work in department A. One employee will be selected at random.

What is the probability that the selected employee works in department A and is not classified as junior level?

- (A) 0.10
- (B) 0.19
- (C) 0.22
- (D) 0.32
- (E) 0.50

14. A one-sample z -test for a proportion at the significance level of $\alpha = 0.10$ will be conducted for the following set of hypotheses.

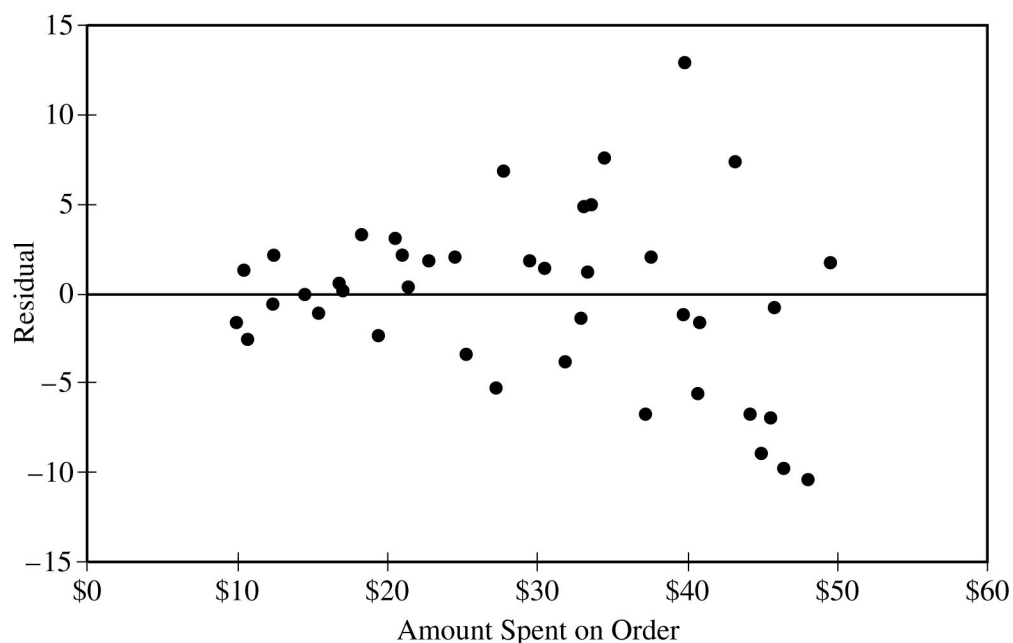
$$H_0 : p = 0.50$$

$$H_a : p < 0.50$$

For which of the following is the probability of a Type II error the greatest?

- (A) A sample size of 200 and a true proportion of 0.45
- (B) A sample size of 200 and a true proportion of 0.40
- (C) A sample size of 200 and a true proportion of 0.35
- (D) A sample size of 100 and a true proportion of 0.45
- (E) A sample size of 100 and a true proportion of 0.35

15. A small-business owner collected data from a random sample on the amount spent on an order and the time it takes to fill the order. The resulting data were used to complete a linear regression analysis of time, in minutes, versus amount spent, in dollars, and to create the following residual plot.



Based on the residual plot, which condition for inference for the slope of the regression line does not appear to be satisfied?

- (A) The true relationship between the number of items and the amount spent on an order is linear.
- (B) For each amount spent on an order, the distribution of time to fill the order is approximately normal.
- (C) The standard deviation of the distribution of time to fill an order is the same for each amount spent on the order.
- (D) The data are collected using a random sample or randomized experiment.
- (E) The sample size is less than 10% of the population size.

16. An athletic trainer wanted to investigate the relationship between an athlete's resting heart rate and the heart rate after exercise. From a sample of 12 athletes, the athletic trainer recorded each athlete's resting heart rate, in beats per minute, and heart rate, in beats per minute, after 5 minutes of moderate exercise. The results of a regression analysis of the data are shown in the following computer output.

Predictor	Coef	SE Coef	T	P
Constant	17.5	29.2	0.60	0.563
Resting Heart Rate	1.786	0.453	3.94	0.003
S = 9.9702		R-Sq = 60.88%		

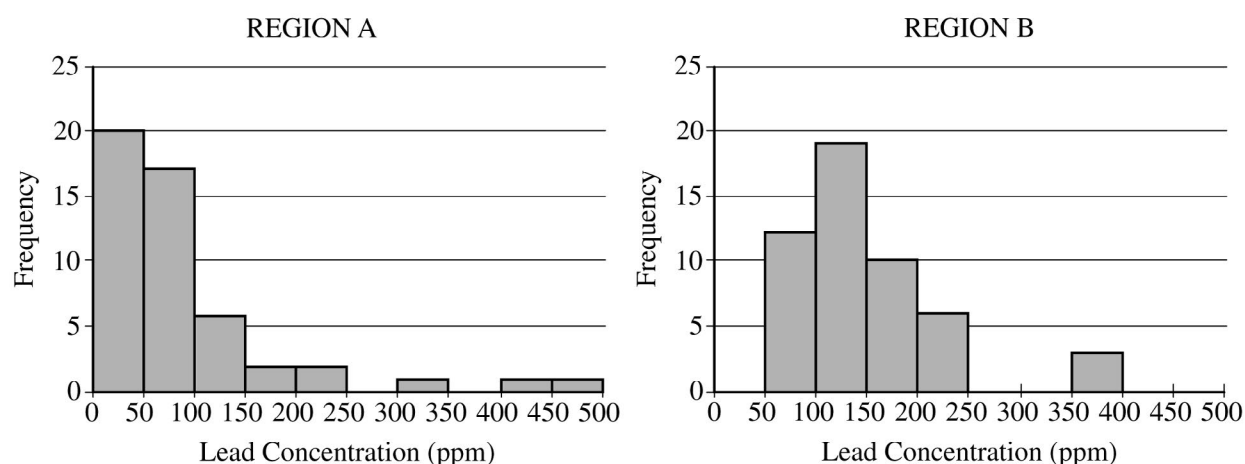
Assume the conditions for inference are met. Which of the following represents a 95 percent confidence for the slope of the population regression line?

- (A) $17.5 \pm 1.96(29.2)$
 (B) $1.786 \pm 1.96(0.453)$
 (C) $1.786 \pm 1.96\left(\frac{0.453}{\sqrt{12}}\right)$
 (D) $1.786 \pm 2.228\left(\frac{0.453}{\sqrt{12}}\right)$
 (E) $1.786 \pm 2.228(0.453)$

Section II: Free-Response

The following are examples of the kinds of free-response questions found on the exam. Note that on the actual AP Exam, there are six free-response questions.

1. A geologist studying lead concentration in soil selected random samples of soil from two regions: region A and region B. The following histograms show the distribution of lead concentration, in parts per million (ppm), for the two samples.



- (A) Write a few sentences comparing the distributions of lead concentration in the two samples.
- (B) To investigate whether the mean lead concentration is different in region A than in region B, the geologist conducted the appropriate test. All conditions for inference are met, and the p -value of the test is 0.007. Based on the p -value, is there convincing statistical evidence, at a level of significance of $\alpha = 0.05$, that there is a difference between the mean lead concentration of region A and the mean lead concentration of region B? Justify your answer.
2. Past records from a certain movie theater indicate that 80 percent of moviegoers make a purchase at the movie theater's snack bar. A random sample of 3 moviegoers from the theater will be selected at random. Let the random variable S represent the number of moviegoers who make a purchase at the snack bar from the 3 selected.
- (A) Complete the following table to create the probability distribution of S .

s	0	1	2	3
$P(s)$	0.008			0.512

- (B) Calculate and interpret the expected value of S .

Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective	Unit
1	A	2.A	UNC-1.H	1
2	D	3.A	UNC-3.B	4
3	D	3.E	VAR-6.K	6
4	B	3.C	UNC-3.R	5
5	B	1.F	VAR-6.D	6
6	C	3.A	VAR-2.B	1
7	B	4.B	DAT-1.H	2
8	E	1.C	DAT-2.C	3
9	D	3.E	VAR-8.M	8
10	D	4.B	UNC-4.S	7
11	D	1.C	VAR-3.A	3
12	A	4.E	DAT-3.F	7
13	A	3.A	VAR-4.D	4
14	D	4.A	UNC-5.C	6
15	C	4.C	VAR-7.L	9
16	E	3.D	UNC-4.AF	9

Free-Response Question	Skill	Learning Objective	Unit
1	2.A, 2.D, 4.E	UNC-1.H, UNC-1.M, UNC-1.O, DAT-3.H	1, 7
2	2.B, 3.A, 3.B, 4.B	UNC-3.B, VAR-5.A, VAR-5.C, UNC-3.C, VAR-5.D	4

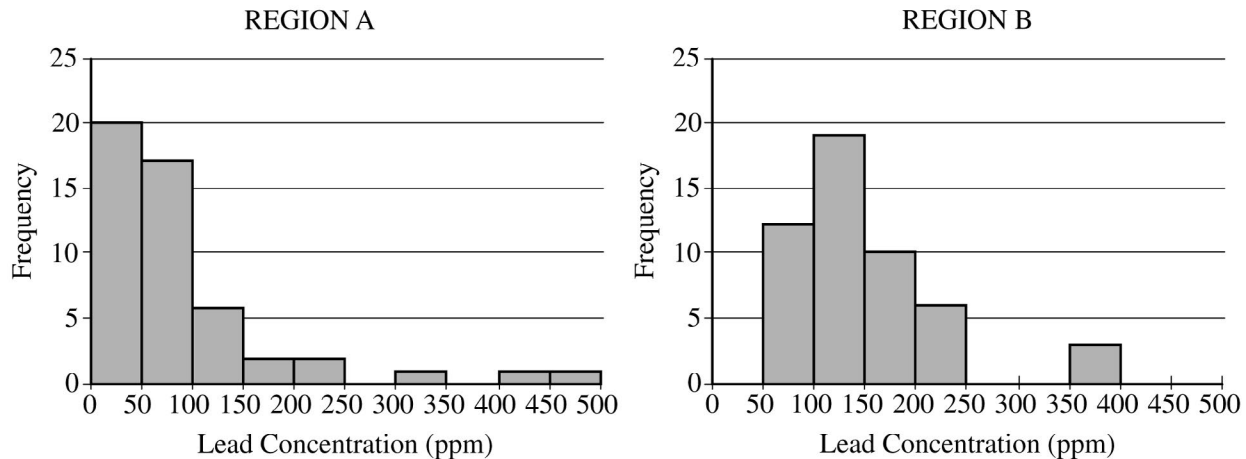
The scoring information for the questions within this course and exam description, along with further exam resources, can be found on the [AP Statistics Exam Page](#) on AP Central.



Scoring Guidelines

Question 1: Focus on Exploring Data

1. A geologist studying lead concentration in soil selected random samples of soil from two regions: region A and region B. The following histograms show the distribution of lead concentration, in parts per million (ppm), for the two samples.



- (A) Write a few sentences comparing the distributions of lead concentration in the two samples.
- (B) To investigate whether the mean lead concentration is different in region A than in region B, the geologist conducted the appropriate test. All conditions for inference are met, and the p -value of the test is 0.007. Based on the p -value, is there convincing statistical evidence, at a level of significance of $\alpha = 0.05$, that there is a difference between the mean lead concentration of region A and the mean lead concentration of region B? Justify your answer.

Scoring Guidelines for Question 1: Focus on Exploring Data

4 points

General Scoring Notes

Each part of the question (indicated by a letter) is initially scored by determining if it meets the criteria for essentially correct, partially correct, or incorrect. The response is then categorized based on the scores assigned to each letter part and awarded a score between 0 and 4 (see the table at the end of each question).

	Model Solution	Scoring
Part (A) 2.A 2.D UNC-1.H UNC-1.M UNC-1.O	The distributions of lead concentration for both Region A and Region B are skewed to the right. The median lead concentration of Region B (between 100 ppm and 150 ppm) is greater than the median lead concentration of Region A (between 50 ppm and 100 ppm). The range of lead concentration of Region A (between 400 ppm and 500 ppm) is greater than the range of lead concentration of Region B (between 250 ppm and 350 ppm)	Essentially correct (E) if the response addresses the following five components: 1. Shape (right skewed) 2. Center (e.g., correct location of mean and/or median) 3. Variability (e.g., correct range and/or IQR) 4. Comparison between regions of at least one of the following: shapes, centers, or variability 5. Context (lead concentration) Partially correct (P) if the response includes only three or four of the five components listed above. Incorrect (I) if the response does not satisfy the criteria for E or P.
Part (B) 4.E DAT-3.H	Because the p-value of 0.007 is less than the level of significance of 0.05, the data provide convincing statistical evidence that there is a difference between the mean lead concentration in the two regions.	Essentially correct (E) if the response provides both of the following: 1. Correct conclusion (Yes, <i>the data provide convincing statistical evidence</i>) in context (<i>lead concentration</i>) 2. Justification based on comparing the p-value to the significance level, α Partially correct (P) Option 1: The response provides both of the following: 1. Correct conclusion (Yes) but without context 2. Justification based on comparing the p-value to the significance level, α Partially correct (P) Option 2: The response provides both of the following: 1. Correct conclusion (Yes) in context (<i>lead concentration</i>) 2. Justification based on the p-value, but without comparison of the p-value to the significance level, α Incorrect (I) if the response does not satisfy the criteria for E or P.

Scoring for Question 1	Score
Complete Response Both parts essentially correct	4
Substantial Response One part essentially correct and one part partially correct	3
Developing response Both parts partially correct OR one part essentially correct and one part incorrect	2
Minimal Response One part partially correct and one part incorrect	1

Question 2: Focus on Probability and Sampling Distributions

2. Past records from a certain movie theater indicate that 80 percent of moviegoers make a purchase at the movie theater's snack bar. A random sample of 3 moviegoers from the theater will be selected at random. Let the random variable S represent the number of moviegoers who make a purchase at the snack bar from the 3 selected.

(A) Complete the following table to create the probability distribution of S .

s	0	1	2	3
$P(s)$	0.008			0.512

(B) Calculate and interpret the expected value of S .

Scoring Guidelines for Question 2: Focus on Probability and Sampling Distributions

4 points

General Scoring Notes

Each part of the question (indicated by a letter) is initially scored by determining if it meets the criteria for essentially correct, partially correct, or incorrect. The response is then categorized based on the scores assigned to each letter part and awarded a score between 0 and 4 (see the table at the end of each question).

Learning Objectives: UNC-3.B VAR-5.A VAR-5.C UNC-3.C VAR-5.D

	Model Solution	Scoring										
Part (A) 2.B 3.A UNC-3.B VAR-5.A	Random variable S has a binomial distribution with $n = 3$ and $p = 0.8$. $P(s = 1) = \binom{3}{1}(0.8)^1(0.2)^2 = 0.096$ $P(s = 2) = \binom{3}{2}(0.8)^2(0.2)^1 = 0.384$ <table><tr><td>s</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>$P(s)$</td><td>0.008</td><td>0.096</td><td>0.384</td><td>0.512</td></tr></table>	s	0	1	2	3	$P(s)$	0.008	0.096	0.384	0.512	Essentially correct (E) if the response includes the following three components. - Shows evidence that the calculation of at least one probability necessary to complete the table is based on a binomial with $n = 3$ and $p = 0.8$, either by using the binomial formula with correct values substituted in, or by using the notation $B(3, 0.8)$, or by using a calculator command, such as <code>binompdf</code> ($n = 3$, $p = 0.8$, $s = 2$). However, an unlabeled calculator command such as <code>binompdf(3, 0.8, 2)</code> does not satisfy component 1. Note: - A normal approximation is not appropriate because the sample size is too small. - Correct calculation of $P(s = 1)$ or $P(s = 2)$. - The probabilities of the probability distribution sum to 1. Partially correct (P) if the response includes only two of the three components. Incorrect (I) if the response does not satisfy the criteria for E or P.
s	0	1	2	3								
$P(s)$	0.008	0.096	0.384	0.512								
Part (B) 3.B 4.B VAR-5.C UNC-3.C VAR-5.D	Calculation, Option 1: The expected value of S is $E(S) = \sum sP(s)$ $= 0(0.008) + 1(0.096) + 2(0.384) + 3(0.512)$ $= 2.4$ Calculation, Option 2: Because S is a binomial random variable, $E(S) = np = 3(0.8) = 2.4$. Interpretation: If the process of randomly selecting 3 moviegoers is repeated many times, the average number of moviegoers who make a purchase at the snack bar will be very close to 2.4.	Essentially correct (E) if the response satisfies the following four components. - Calculates the expected value correctly. A correct expected value calculation that is consistent with incorrect probabilities reported in part (a) satisfies component 1. If the expected value calculation is rounded to an integer, the response does not satisfy component 1. - Interpretation—includes reference to long run, many samples or repeated sampling. - Interpretation—includes reference of average. - Context. Partially correct (P) if the response includes only three of the four components. Incorrect (I) if the response does not satisfy the criteria for E or P.										

Scoring for question 2	Score
Complete Response Both parts essentially correct	4
Substantial Response One part essentially correct and one part partially correct	3
Developing response Both parts partially correct OR one part essentially correct and one part incorrect	2
Minimal Response One part partially correct and one part incorrect	1